

Special Topics in Astrodynamics ae4889

Assignment tasks & deliverables

The assignment for the course Special Topics in Astrodynamics ae4889 is phrased in the form of a mock research proposal with clear tasks to be performed by you. The research proposal is about **the use of hybrid chemical/solar-sail propulsion for establishing a transfer from Earth to a parking orbit around Mars in support of a future human, cargo, or scientific mission to Mars**. Each student will be given their own case to examine (e.g., unique solar-sail parameters) so that the work from all students combined forms a large parametric analysis that will help answering the research questions posed in the research proposal.

Mock research proposal

Title

MIMOSA – Mission to Mars with a solar sail

Background

Mars is located in the habitable zone of the Sun and is hypothesised to have harboured microbial life at some point during its history. It offers clues on the existence of life beyond the Earth, how planets form and evolve, and the exploration of Mars ultimately helps us understand our own origin. The robotic exploration of Mars including orbiters, surface rovers, and aerial vehicles have thus far offered unprecedented details on its surface and interior.

Mars is the most likely place that humans can visit beyond the Moon and considered as the final frontier in human exploration [1]. The ability to live on another planet, see Figure 1, reduces the existential risk that humans may face in the future due to, for example, a catastrophic asteroid impact and ongoing climate change. Furthermore, technological advancements and research on human biology that follow from humans visiting Mars could also enhance life on Earth, as demonstrated during the Apollo programme to the Moon.

On a more practical level, robotic exploration of Mars is significantly reliant on the delayed communication between the Earth and the spacecraft on and around Mars. The autonomous capabilities of rovers and aerial vehicles are also limited, resulting in extremely slow exploration missions. Indeed, human exploration of Mars allows real-time decision making, complex sample selection, and the geological knowledge of humans can reduce the exploration duration from months or years to just a few days, offering a significant boost to exploration campaigns [2].

However, human exploration of other planets brings significant challenges to be addressed. The backbone of human exploration is the basic infrastructure necessary prior to crew arrival and its building blocks. First and foremost, a long-term crew habitat and associated infrastructure are imperative. This includes radiation protected, closed-loop life-support systems, energy (e.g., solar or nuclear) and resource utilisation using in-situ materials and navigation systems for positioning and systems to communicate with the Earth [2].



Figure 1 An image depicting a conceptual crewed exploration of Mars (Credits: JPL)

Building such an infrastructure in the short term requires a large capacity of space transport systems that can be launched in rapid cadence. While launchers such as SpaceX's Starship could offer such capacity, reusability and cadence, the supply of infrastructure to the surface of Mars is still limited by the propellant capability and orbital dynamics itself. For the former (propellant capability), the concept of orbital depots is proposed as "refuelling stops" in the interplanetary space between Earth and Mars [3], yet these are also limited by their capacity and bring additional challenges such as in-space rendezvous and docking. The latter (orbital dynamics) is a greater problem, which restricts the launch and transfer windows between Earth and Mars. A typical Hohmann-like transfer between the Earth and Mars can only be achieved approximately every two years. This natural restriction on transfer windows significantly hinders the infrastructure development on Mars.

Nevertheless, the natural environment and dynamics within the solar system also offer new pathways that can help overcome these limitations. To that end, the gravitational landscape characterised by the Sun and Mars can be exploited to identify natural, low-energy trajectories to enable transfers between the vicinity of the Earth and the vicinity of Mars. The gravitational landscape can be formulated by the circular restricted three-body problem, which constitutes two gravitational objects (i.e., Sun and Mars) orbiting each other in a circular orbit with the third object (spacecraft) moving under their influence.

The possibilities offered by this environment can be enhanced by utilising the pressure of solar radiation in a method called solar sailing. Solar sailing requires no onboard propellant hence offering, in theory, "limitless" transfer opportunities. But the low thrust from the sail results in longer escape from the Earth's gravity well and insertion into a Mars-bound orbit. This can be compensated with a hybrid system where a chemical thruster can be combined to provide the necessary escape and insertion speed. This hybrid concept using natural dynamics (in particular, the 1st and 2nd Lagrange points of the Sun/Mars system and their manifolds), solar sailing and chemical propulsion, see Figure 2, can overcome transfer window limitations while reducing the overall propellant cost.

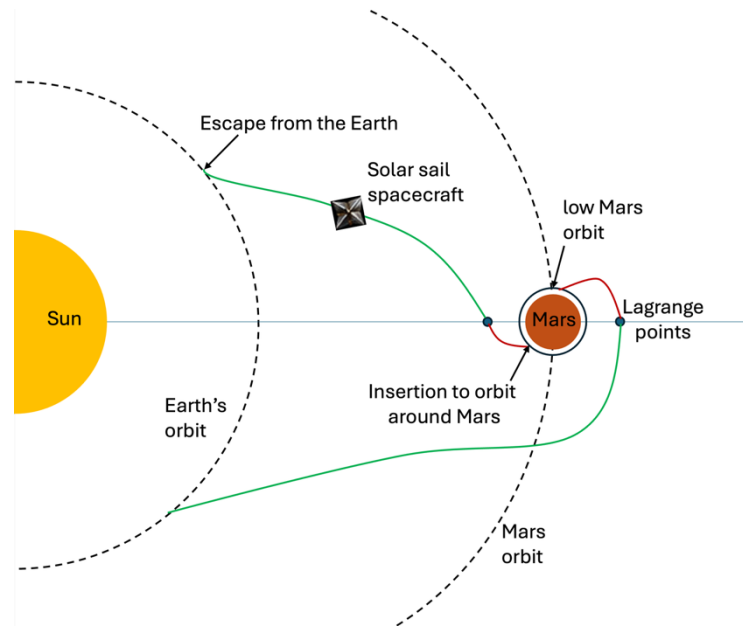


Figure 2 Proposed concept for the Earth-Mars transfer

Research objective and research question(s)

While some previous studies have looked into natural ballistic capture for transfers from Earth to Mars [4], natural transfer trajectories between individual periodic orbits in the Sun-Earth and the Sun-Mars systems [5, 6], and transfers to Mars-bound orbits [7], an end-to-end scenario that uses a blend of solar sailing and chemical propulsion in the three-body problem of the Sun-Mars system is currently not available. The proposed research objective therefore is:

To analyse the capabilities of and requirements for hybrid chemical/solar-sail propulsion to transfer a human, cargo, or scientific payload from Earth to a parking orbit around Mars.

By achieving this research objective, the following main research question may be answered:

Can the hybrid chemical/solar-sail propulsion architecture aid in establishing a transfer from Earth to a parking orbit around Mars, thereby contributing to the human and/or scientific exploration of Mars, while simultaneously presenting an excellent use case for solar-sail propulsion?

More detailed sub-research questions that may be answered are:

- 1) Do the colinear Lagrange points of the Sun-Mars system offer a gateway and thereby a valid design approach for a solar-sail transfer from Earth to a Martian parking orbit?
- 2) Which colinear Sun-Mars Lagrange point (L_1 or L_2) forms the best gateway for a trajectory to Mars, where best is defined as the minimum required chemical insertion ΔV s at Earth and the Martian parking orbit?
- 3) What is the effect of
 - 1) the solar-sail capabilities (in terms of area-to-mass ratio)
 - 2) the altitude of the Martian parking orbit
 on the transfer performance in terms of the required insertion ΔV s at Earth orbit and the Martian parking orbit?
- 4) What solar-sail steering strategy allows for optimal transfer performance?

Workplan

To answer the research questions posed above and fulfil the stated research objective, the work packages as illustrated in Figure 3 have been defined.

WP1: Locate the Lagrange points – the classical co-linear Lagrange points of the Sun-Earth and Sun-Mars circular restricted three-body problems (CR3BPs) are computed.

WP2: Compute the artificial Lagrange points – a low-fidelity solar-sail acceleration model is implemented, and the artificial Lagrange points that result in the solar-sail augmented CR3BP are explored.

WP3: Assess the stability of the classical and artificial Lagrange points – The stability of the Lagrange points obtained under WP1 and WP2 are computed and assessed.

WP4: Create solar-sail invariant manifolds and assess their utility for a Mars exploration mission scenario – the (un)stable manifolds emanating from the classical Lagrange points are evaluated, their use in designing a **Mars exploration mission scenario** is examined, and solar-sail propulsion is added to create a transfer and minimise the total ΔV required for the mission.

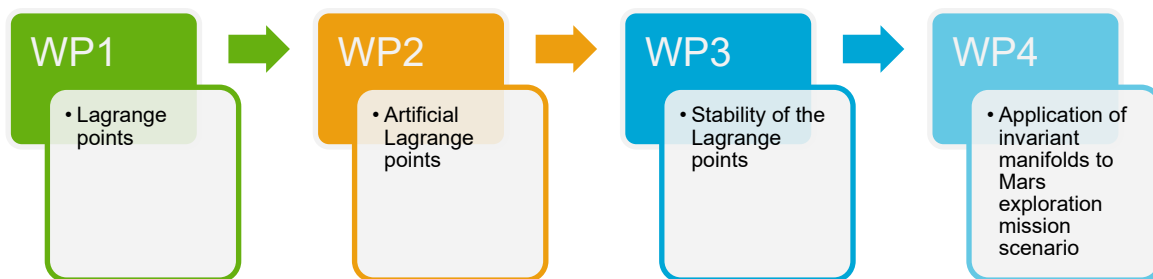


Figure 3 Work-packages for the research proposal "MIMOSA".

Deliverables

The final deliverables are:

1) **A report.** This report should:

- Include a title page with your name, student number, and details of **your personal problem specifications** using tables like the ones below (the example table contains dummy data). **Also indicate on the title page who you collaborated with (if anyone).**

Student name	Jane Doe
Student number	1111111

Personal problem specification	
Parameter	Value
Parameter 1	x
Parameter 2	x
Parameter 3	x

Collaborated with	John Doe
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- Include only the answers to the tasks outlined in this mock research proposal (derivations should not be included).
 - Clearly indicate which part of your report belongs to which (sub)task, i.e., **provide your answer per (sub)task**; it should be clear what part of your report belongs to which (sub)task.
 - **Follow the format instructions provided for each task. Not complying with the specified format will have an effect on your assessment.**
 - Include a list of indexed references.
- 2) **Your code.** Upload your code as a clearly identifiable set of files in a new, *private* GitHub repository and include the link to the repository in the report. Invite the responsible instructor and other teaching staff involved in the course to this private GitHub repository so they can access your code. **The GitHub handles of the instructors are celikonur and mjheiligers.**

Some final important remarks

Regarding the technical tasks

- The values for **your personal problem specification** can be found on Brightspace under “Assignment”.
- Values for common astrodynamics constants and parameters can be found in the Appendix and **need to be used** in the execution of the tasks detailed below.

Deadlines

- You are offered a chance (**not compulsory**) to receive **formative feedback** on your tasks for WP1 and WP2 if you submit the answers to these tasks by **Friday, 8 May 2026 23:59 (midnight)**.
- Finally, note that the **deadline for submission of the full assignment is set on Friday, 12 June 2026 at 23:59 (midnight)**. Submission after this deadline will result in a **one point deduction per 24 hours** that your submission is late.

WP1 Sun-Earth and Sun-Mars Lagrange points

Task 1.1a Compute the coordinates of the Sun-Earth and Sun-Mars colinear Lagrange points *in your personal problem specification* using Newton's method.

Format: use a table like the one in Table 1 to provide the coordinates (x_i, y_i, z_i) ¹ of the Lagrange points assigned to you in the respective Sun-planet synodic reference frame $S(x_i, y_i, z_i)$ and in dimensionless form. Please replace "L_j" with the Lagrange point assigned to you in your personal problem specification.

Table 1 Dimensionless coordinates (x_i, y_i, z_i) of the Sun-Earth and Sun-Mars colinear Lagrange points in their respective synodic reference frames.¹

Lagrange point	x_i	y_i	z_i
Sun-Earth - L _j			
Sun-Mars - L _j			

Task 1.1b Verify the correctness of the coordinates obtained under Task 1.1a.

Format: use a maximum of 50 words (excluding any references used) to describe your method of verification and the outcome and conclusion of the verification. Indicate the number of words used.

Task 1.2 Comment on the difference in distance between the Lagrange point and the planet for the Sun-Earth and Sun-Mars Lagrange points found under Task 1.1a. Explain the difference by considering **all** acceleration vectors acting in the CR3BP.

Format: use a maximum of 100 words (excluding any references used) to explain the different distances. Indicate the number of words used.

¹Subscript $i=[E,M]$ with E and M referring to Earth and Mars, respectively

WP2 Artificial Lagrange points

Task 2.1a We now want to equip the spacecraft with a solar sail; this will allow a shift in the location of the Lagrange points. Assume we want to shift both Lagrange points found in Task 1.1a by a certain amount; and we want this shift to occur along the line connecting the Sun and the Lagrange point *in the direction of the Sun*. The amount by which the Lagrange points shall be shifted can be found *in your personal problem specification*.

- Compute the required solar-sail performance, i.e., the solar-sail lightness number, to achieve this shift. Assume an ideal performance of the sail.
- Comment on the near-term feasibility of the required solar-sail performance and justify your reasoning.

Format:

- Use a table like the one in Table 2 to provide the required solar-sail lightness number. Please replace “ L_j ” with the Lagrange point assigned to you in your personal problem specification.
- For your comment and justification on the near-term feasibility, use a maximum of 50 words (excluding any references used). Indicate the number of words used.

Table 2 Required lightness number to achieve the shift in the Lagrange points as specified in your personal problem specification.

	Shift in Lagrange point along the Sun-planet line ²	Required lightness number, β
Sun-Earth - L_j		
Sun-Mars - L_j		

Task 2.1b Compute and explain the required orientation of the sail by specifying the orientation of the sail normal vector. Explain the obtained sail orientation by considering **all** acceleration vectors acting in the solar-sail augmented CR3BP.

Format:

- Use a table like Table 3 below to provide the components of the normal vector, $\mathbf{n}_i = [n_{x,i} \ n_{y,i} \ n_{z,i}]^T$, in the Sun-planet synodic reference frame.¹
- Use a schematic of all acceleration vectors acting in the solar-sail augmented CR3BP and a maximum of 100 words (excluding any references used) to explain the obtained sail orientation.

Table 3 Required sail orientation to achieve the required shift in the Lagrange points of Task 2.1a.

	$n_{x,i}$	$n_{y,i}$	$n_{z,i}$
Sun-Earth - L_j			
Sun-Mars - L_j			

² Please provide the shift as stated in your personal problem specification.

Task 2.2 Verify the correctness of the lightness number **and** sail orientation obtained under Tasks 2.1a and 2.1b.

Format: Use a maximum of 100 words (excluding any used references) to describe your method of verification and outcome and conclusion of the verification. Indicate the number of words used.

Task 2.3 Compute the required size of the solar sail (in terms of solar-sail side length) for a 100-kg spacecraft and put this solar-sail size in perspective with current solar-sail technology.

Format: Use a table like the one in Table 4 to provide the required size of the solar sail, through the required side-length (area = side-length x side-length) in metres, for the shift in each of your two artificial Lagrange points found under Task 2.1a.

Table 4 Required solar-sail side-length to achieve the required shift in the Lagrange points of Task 2.1a.

Required solar-sail side-length [m]	
Sun-Earth Lagrange point	
Sun-Mars Lagrange point	

Task 2.4 Conduct a sensitivity analysis on the required solar-sail lightness number when changing the shift in the Lagrange points. Compute the required lightness number by stepwise increasing the shift until the shift is **twice** that specified *in your personal problem specification*. Choose an appropriate step size to present your results. Discuss the obtained trend in the coordinates of the artificial Lagrange points by considering the changes in the accelerations acting on the solar sail when varying the shift.

Format: Plot the solar-sail lightness number as a function of the x_i -coordinate of the Sun-Earth and Sun-Mars artificial Lagrange points. Use axes labels and a legend to make the figure legible. For your discussion on the trend observed, use a maximum of 100 words (excluding any references used). Indicate the number of words used.

WP3 Stability of the (artificial) Lagrange points

Task 3.1a Compute the stability of the classical Lagrange points (i.e., the ones found under Task 1.1a) and the artificial Lagrange points (i.e., the ones found under Task 2.1a) through an eigenvalue analysis of the Jacobian of the linearised system.

Format: use a table like the one in Table 5 to provide the eigenvalues, $\lambda_1 \dots \lambda_4$ of the classical and artificial Lagrange points.

Table 5 Eigenvalues of the classical and artificial Lagrange points.

Lagrange point	λ_1	λ_2	λ_3	λ_4
Sun-Earth Classical (Task 1.1a)				
Sun-Earth Artificial (Task 2.1a)				
Sun-Mars Classical (Task 1.1a)				
Sun-Mars Artificial (Task 2.1a)				

Task 3.1b Comment on the stability of all Lagrange points as found in Table 5 and on how the solar-sail acceleration changes the stability of the Lagrange points.

Format: Use a maximum of 100 words (excluding any references used). Indicate the number of words used.

Task 3.2 Verify the correctness of the eigenvalues of the Lagrange points as found in Table 5.

Format: use a maximum of 100 words (excluding any used references) to describe your method of verification and outcome and conclusion of the verification. Indicate the number of words used.

WP4 Invariant manifolds and application to Mars exploration mission

Task 4.1 Compute and plot the unstable and stable manifolds of **the classical Sun-Mars Lagrange point** in your personal problem specification. Use absolute and relative integration tolerances of 10^{-12} , a propagation time of 5 (**Earth**) years, and a dimensionless perturbation along the eigenvectors of 10^{-5} .

Format: Plot the manifold trajectories on a suitable grid size with an aspect ratio such that the data units are the same in every direction. In your plots, use different colours and/or line styles to make a clear distinction between the unstable and stable manifolds. Use axes labels and a legend to make the figure legible. Provide:

- *System-wide view of the (x,y)-projection of the (un)stable manifolds so that the full length of the manifolds is clearly visible.*
- *Close-up view of the (x,y)-projection of the (un)stable manifolds so that the manifold structure close to Mars is clearly visible. Add the Martian parking orbit to this close-up view, i.e., a circular orbit around Mars with an altitude as specified in your personal problem specification.*

Task 4.2a Comment on the use of the classical manifolds to establish a feasible transfer between Earth and the Martian parking orbit plotted in Task 4.1. Here, feasibility of the transfer is defined when the manifolds³ intersect

- Earth's orbit, i.e., a circular orbit at 1 au from the Sun (this assumption holds throughout the rest of the assignment)
- The Martian parking orbit (again, a circular orbit around Mars with an altitude as specified in your personal problem specification).

Format: Use a maximum of 50 words (excluding any references used) to comment on the transfer feasibility. Indicate the number of words used.

Task 4.2b If, in Task 4.2a, you have identified an intersection (either at Earth, at Mars, or both), compute the ΔV to be inserted into Earth's orbit, ΔV_E (the orbit as defined in Task 4.2a) and/or the ΔV to be inserted into the Martian parking orbit, ΔV_M (again, the orbit as defined in Task 4.2a) and the total ΔV , $\Delta V_{TOT} = \Delta V_E + \Delta V_M$.

Format:

- *Use a table like the one in Table 6 to provide the values for the three ΔV s as defined above. Provide all ΔV s in m/s.*
- *If you found an intersection at Mars, provide a close-up view of the transfer from the Lagrange point to the parking orbit. In your figure, provide the Lagrange point, the transfer up to the intersection with the parking orbit, the parking orbit and mark the exact location of the intersection. In your plot, use different colours and/or line styles to make a clear distinction between the different elements in the figure. Use axes labels and a legend to make the figure legible.*

³ Note – think about which manifold (i.e. the stable or unstable ones) are needed for each part of the transfer from Earth to Mars.

Table 6 ΔV s required to establish a natural transfer between Earth and Mars.

ΔV_E at Earth's orbit [m/s]	ΔV_M at Martian parking orbit [m/s]	ΔV_{TOT} [m/s]

Task 4.3 Again propagate the manifolds, but now add an ideal solar-sail acceleration to the dynamics and assess the feasibility of your assigned transfer.

For the solar-sail acceleration:

- Use the solar-sail lightness number as specified in your personal problem specification.
- Assume cone angles of $\alpha \in [-90 : 2.5 : 90]$ deg, i.e., a range spanning between -90 deg and +90 deg with a step size of 2.5 deg.
- **Assume a clock angle such that the sail normal vector is contained in the (x,y)-plane and the transfer remains planar.**

For computing the solar-sail augmented manifolds:

- Start propagating from **the classical Sun-Mars Lagrange point**.
- Use the initial condition of the manifolds as indicated in Table 7 for each leg of the transfer, where leg 1 refers to the transfer “Earth $\rightarrow$ Lagrange point” and leg 2 refers to the transfer “Lagrange point $\rightarrow$ Martian parking orbit”.
- Use absolute and relative integration tolerances of 10^{-12} , and a maximum propagation time of 5 (**Earth**) years.

Table 7 Initial condition to be used to generate the solar-sail augmented manifolds.

Your Sun-Mars Lagrange point	Leg 1 Earth $\rightarrow$ Lagrange point	Leg 2 Lagrange point $\rightarrow$ Mars parking orbit
L₁	Stable left	Unstable right
L₂	Stable right	Unstable left

For each cone angle⁴, assess the feasibility and cost of your assigned transfer by computing the ΔV s to be inserted into Earth's orbit (ΔV_E , as defined in Task 4.2b) and/or the Martian parking orbit (ΔV_M , again, as defined in Task 4.2b) and the total ΔV , ΔV_{TOT} .

⁴ Note – A different cone angle can be adopted in the two different legs of the transfer (Earth $\rightarrow$ Lagrange point and Lagrange point $\rightarrow$ Martian parking orbit)

Format:

- Provide a plot of the three ΔV s (ΔV_E , ΔV_M , and ΔV_{TOT}) as a function of the cone angle. Use axes labels and a legend to make the figure legible.
- Use a table like the one in Table 8 to provide the cone angles⁴ for which the smallest total ΔV , ΔV_{TOT} , is obtained and the accompanying transfer time along the Earth-bound and Mars-bound trajectories. Provide these transfer times in **Earth** years.
- Provide two plots of the transfer for which the smallest total ΔV , ΔV_{TOT} , is obtained. One plot should provide a system-wide view, the other plot a close-up view, similar to the plots in Task 4. Add the orbit of the Earth and the Martian parking orbit to the plots. Use an aspect ratio such that the data units are the same in every direction. Use axes labels and a legend to make the figure legible.

Table 8 Details of the solar-sail Earth-Mars transfer with the smallest total ΔV , ΔV_{TOT} .

Leg 1 Optimal cone angle [deg]	Leg 2 Optimal cone angle [deg]	ΔV_E at Earth's orbit [m/s]	ΔV_M at Martian parking orbit [m/s]	ΔV_{TOT} [m/s]	Transfer time along Earth- bound trajectory [Earth years]	Transfer time along Mars- bound trajectory [Earth years]

Task 4.4 Meet with a fellow student who completed Task 4.3 and was assigned a different parameter set (i.e., personal problem specification) than you. Compare your results and discuss the difference in the results (i.e., all quantities in Table 8) in light of the different parameter set. What do you observe and what can you conclude?

Format: Use a maximum of 200 words (excluding any used references) to comment on the different transfer performance in relation to the parameter set. Indicate the number of words used.

Task 4.5 Given that, on 1 January 2030, the Earth is *ahead* of Mars by 122.35 deg in their respective heliocentric orbits, compute the earliest possible launch date for your Earth-to-Mars transfer.

Task 4.6 Provide your view on possible human, cargo, and scientific applications of the solar-sail transfer you found under Task 4.4. What challenges do you foresee with each of these applications? Support, where possible, your view with citations to relevant sources.

Format: Use a maximum of 200 words to provide your view on possible mission applications.

Assessment

For the assessment of your report, a rubric with the following criteria will be used:

1. Scientific soundness
 - a. Correctness of the answers to the tasks defined in the research proposal
 - b. Verification of answers against test data / validation against scientific literature
 - c. Discussion supported by relevant references from scientific literature and publications
 - i. Make good use of guidance, materials, and references provided; find additional, relevant references
 - d. Free of scientific errors
2. Clarity of discussion and logical, scientific argumentation
3. Scientific writing & reporting. For example:
 - a. Correct rounding of final answers
 - b. Declaration and unique use of variables, if applicable
 - c. Legibility of figures (axis labels, legend)
 - d. Cross-referencing of figures and tables
 - e. Impartial and unambiguous language
4. Report includes all elements described under “Deliverables”

Appendix - Table of constants and parameter values

Parameter	Value
Astronomical unit	149,597,870.7 km
Earth's sidereal year	$365.256 \times 24 \times 3600$ s
Gravitational constant of Mars	0.042828×10^6 km ³ /s ²
Gravitational constant of the Sun	$132,712 \times 10^6$ km ³ /s ²
Mass ratio of Sun-Earth system ⁵	3.0542×10^{-6}
Mass ratio of Sun-Mars system ⁵	$3.227154996101724 \times 10^{-7}$
Time unit of Sun-Mars system	8,253,622 s
Length unit of Sun-Mars system	208,321,282 km

⁵ NASA, Three-Body Periodic Orbits, https://ssd.jpl.nasa.gov/tools/periodic_orbits.html#/periodic, Accessed 1 May 2026

References

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- [2] NASA, 2023. Science Objectives for Human Exploration of Mars Workshop Report. Available via: NASA Technical Reports Server (NTRS). Available at <https://ntrs.nasa.gov/api/citations/20230001230/downloads/Science%20Objectives%20for%20Human%20Exploration%20of%20Mars%20Workshop%20Report%20%20v8.docx.pdf> (Accessed on 22 April 2026)
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